Bilkent University
Department of Electrical and Electronics Engineering
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Arbitrary Waveform Generator Using Clocking Wizard IP

Abstract: The objective of this laboratory session was to design and implement an arbitrary digital waveform generator on a Basys 3 FPGA board using VHDL. A Clocking Wizard IP was employed within a Vivado block design to generate a clean 100 MHz reference clock. A four-phase finite state machine, operating within a single synchronous process, produced a repeating asymmetric pulse train with durations of 40 µs, 90 µs, 60 µs, and 20 µs (total period 210 µs). Two physical switches provided real-time waveform modification: one selecting between two distinct timing patterns, and the other doubling the output speed by halving all phase durations. Behavioral simulation confirmed zero-error timing accuracy across all four switch configurations, and hardware verification via oscilloscope validated the generated waveforms against theoretical predictions.

1. Introduction

Arbitrary waveform generation is a fundamental capability in digital systems, enabling the production of precisely timed control signals for applications ranging from motor control to communication protocols. Unlike fixed-frequency clock division, arbitrary waveform generation demands programmable timing for each segment of the output signal. In this laboratory exercise, the Basys 3 FPGA was used to synthesize a repeating four-phase digital waveform whose timing parameters could be altered in real time through physical switch inputs. A Xilinx Clocking Wizard IP was integrated via a block design to provide a stable, jitter-reduced clock source. The entire waveform generation logic was implemented within a single

synchronous VHDL process, adhering to best practices for avoiding glitch-prone combinational outputs and clock domain crossing issues.

2. Theory

2.1. Clock-Based Timing and Counter Arithmetic

The Clocking Wizard IP receives the 100 MHz board oscillator and outputs a phase-locked 100 MHz clock with reduced jitter. At this frequency, each clock period is exactly 10 ns. Generating a precise time interval therefore reduces to counting a specific number of clock cycles:

$$t_{output} = N \times 10\,ns$$

where N is the number of clock ticks. For example, a 40 µs HIGH pulse requires N = 40,000 ns / 10 ns = 4,000 ticks. Because all target durations in this design are exact multiples of 10 ns, there is zero quantization error in the generated waveform, a critical requirement emphasized in the laboratory manual.

2.2. Finite State Machine for Waveform Synthesis

A four-state Moore-type finite state machine governs the output. Each state defines both the output level (HIGH or LOW) and the duration for which that level is held. The state transitions are:

State 0 (HIGH, t1) → State 1 (LOW, t2) → State 2 (HIGH, t3) → State 3 (LOW, t4) → State 0

A single integer counter tracks the elapsed ticks within each state. When the counter reaches the target duration minus one (accounting for zero-indexed counting), it resets to zero and the machine advances to the next state. This minus-one comparison is essential: counting from 0 to N−1 yields exactly N clock edges, whereas counting from 0 to N would produce N+1 edges, introducing a 10 ns timing error.

2.3. Clocking Wizard and Block Design

The Xilinx Clocking Wizard IP wraps an MMCM (Mixed-Mode Clock Manager) primitive within the Artix-7 FPGA. It accepts the raw board oscillator and produces a frequency-synthesized output through a phase-locked loop. Although the output frequency in this design matches the input (100 MHz), the PLL filtering removes clock jitter present in the raw oscillator signal. The block design integrates the Clocking Wizard with the waveform generator RTL module, and Vivado automatically generates the top-level HDL wrapper.

3. Methodology

3.1. Equipment and Components

Hardware: Digilent Basys 3 Artix-7 FPGA Board (XC7A35TCPG236-1), digital oscilloscope with probes. Software: Xilinx Vivado Design Suite. I/O Peripherals: 2 physical slide switches (waveform control), Pmod JA header Pin 1 (waveform output), 2 LEDs (switch state indicators), center push button (asynchronous reset), 100 MHz on-board oscillator.

3.2. Design Architecture

The system consists of two components connected within a Vivado IP Integrator block design:

1. Clocking Wizard (clk_wiz_0): Receives the 100 MHz board clock on clk_in1 and outputs a clean 100 MHz signal on clk_out1. The PLL reset input is tied to a Constant IP outputting logic 0, ensuring the clock infrastructure is never inadvertently disrupted by the user reset button.
2. Waveform Generator (waveform_gen_0): Contains a single synchronous process sensitive only to the Clocking Wizard output clock and an asynchronous reset. Internally, the process performs three operations on every rising clock edge: (a) reads the two switch inputs and loads the corresponding set of four phase-duration constants into registers s_t1 through s_t4; (b) increments a phase counter and evaluates it against the active duration to determine state transitions; (c) drives the output signal HIGH in states 0 and 2, and LOW in states 1 and 3.

The switch mapping provides four waveform configurations:

sw[1:0]	Pattern	Speed	Phase Durations (H/L/H/L)	Period
00	A	1x	40 / 90 / 60 / 20 µs	210 µs
01	B	1x	55 / 35 / 45 / 75 µs	210 µs
10	A	2x	20 / 45 / 30 / 10 µs	105 µs
11	B	2x	27.5 / 17.5 / 22.5 / 37.5 µs	105 µs

3.3. Synthesis and Physical Mapping

An XDC constraints file mapped the block design wrapper ports to the physical Basys 3 pins: the board oscillator to package pin W5 (LVCMOS33), switches sw(0) and sw(1) to pins V17 and V16, the waveform output to Pmod JA Pin 1 (package pin J1), LEDs to pins U16 and E19, and the reset button to pin U18. Port names in the constraints file were verified against the auto-generated HDL wrapper to prevent silent mapping failures.

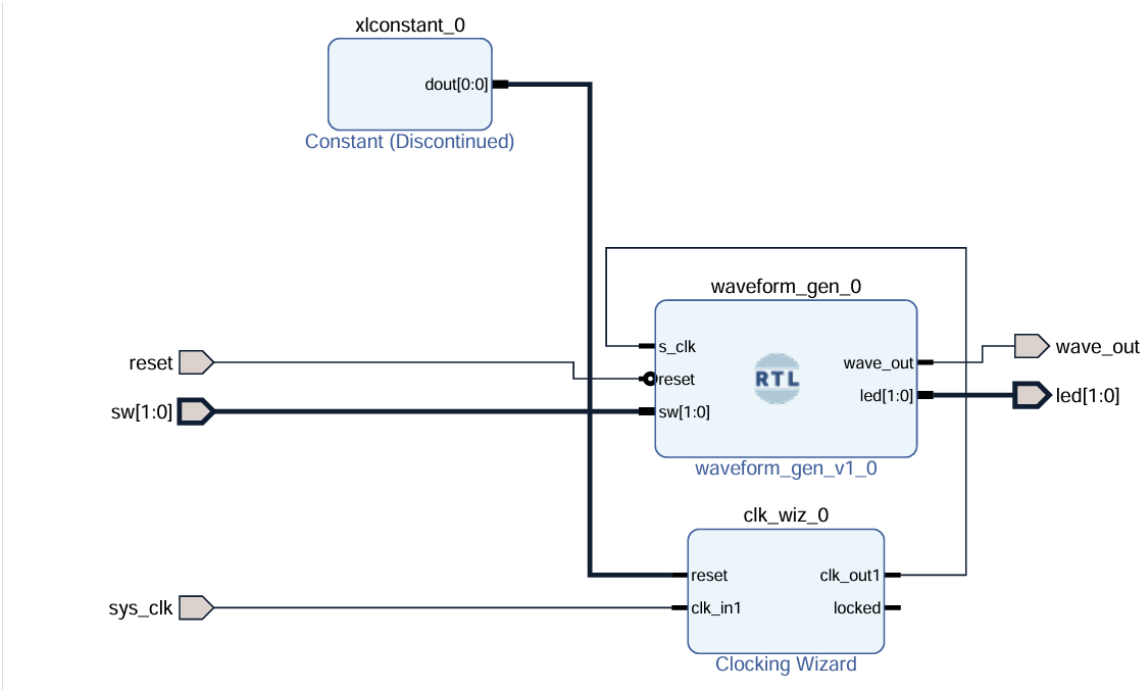


Figure 1: Vivado block design showing Clocking Wizard, Constant IP, and waveform_gen module with all connections and external ports labeled.

4. Results and Analysis

4.1. Behavioral Simulation Verification

A testbench was written to verify all four switch configurations. The testbench generated a 100 MHz clock and sequentially applied each switch combination, holding each for a duration sufficient to observe at least two complete waveform periods. The total simulation time was approximately 1600 µs.

Using waveform markers in the Vivado simulation viewer, the following timing measurements were recorded for the default configuration (sw = "00"):

Phase	Expected Duration	Measured Duration	Error
1st HIGH (State 0)	40.000 µs	40.000 µs	0 ns
1st LOW (State 1)	90.000 µs	90.000 µs	0 ns
2nd HIGH (State 2)	60.000 µs	60.000 µs	0 ns
2nd LOW (State 3)	20.000 µs	20.000 µs	0 ns
Full Period	210.000 µs	210.000 µs	0 ns

The 2x speed configuration (sw = "10") was also verified, confirming a halved period of 105.000 µs with each individual phase duration exactly half of the 1x values.

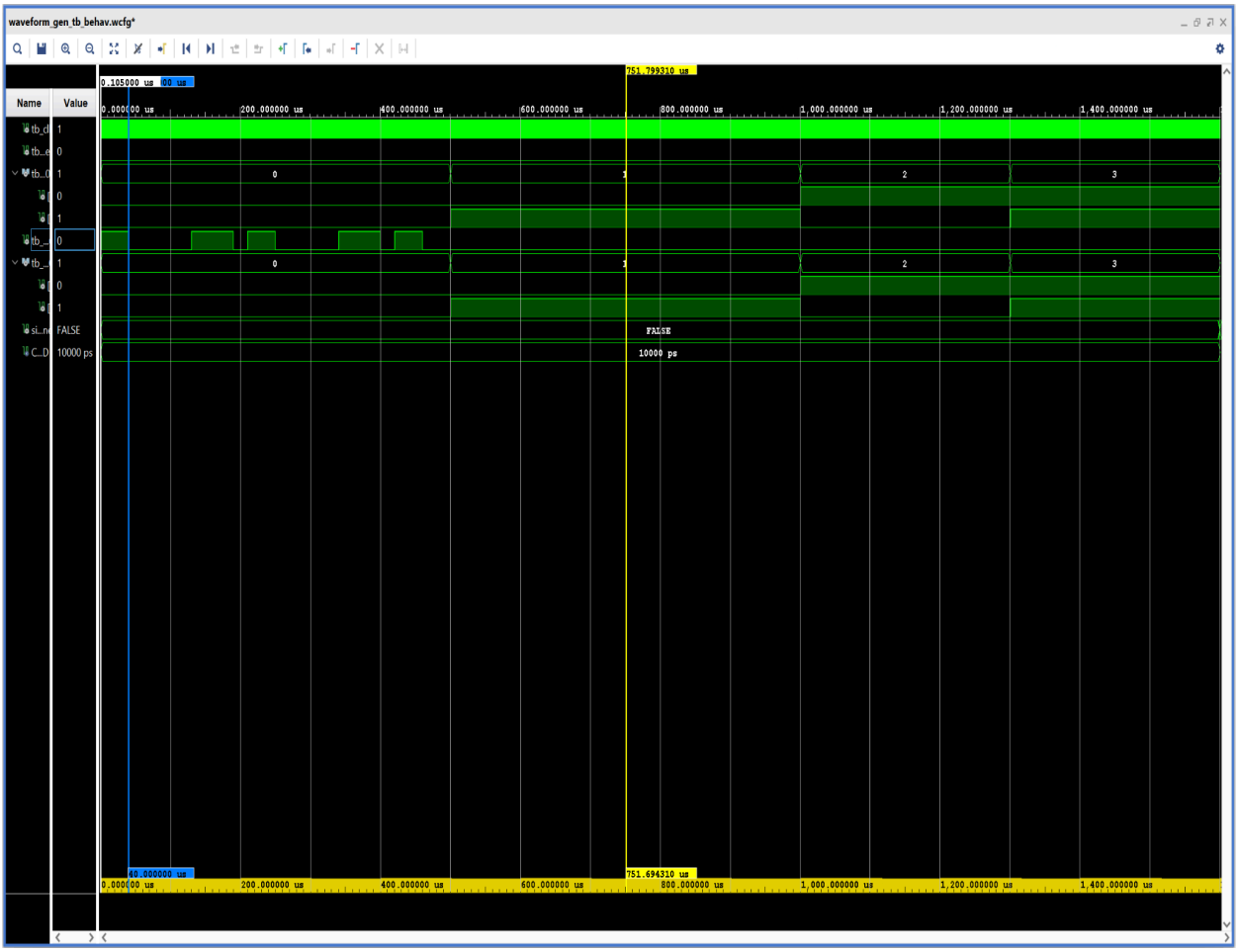


Figure 2: Simulation waveform for sw="00" (Pattern A, 1x speed) showing tb_wave_out with two markers measuring the 40 µs HIGH pulse.

4.2. Hardware Verification

Upon programming the Basys 3 via the Hardware Manager, the waveform output on Pmod JA Pin 1 was probed using a digital oscilloscope. Channel 1 was connected to JA Pin 1 with the ground clip attached to the Pmod GND pin. The oscilloscope was configured with a timebase of 70 µs/div, 500 mV/div vertical scale, and rising-edge trigger at approximately 1 V.

The default pattern (sw = "00") displayed the expected asymmetric pulse train with clearly distinguishable 40 µs and 60 µs HIGH pulses separated by 90 µs and 20 µs LOW intervals. Toggling sw(0) produced a visible change in the pulse shape, confirming pattern switching. Toggling sw(1) compressed the entire waveform horizontally, confirming the 2x speed mode. LED indicators on the board correctly mirrored the switch states throughout testing.

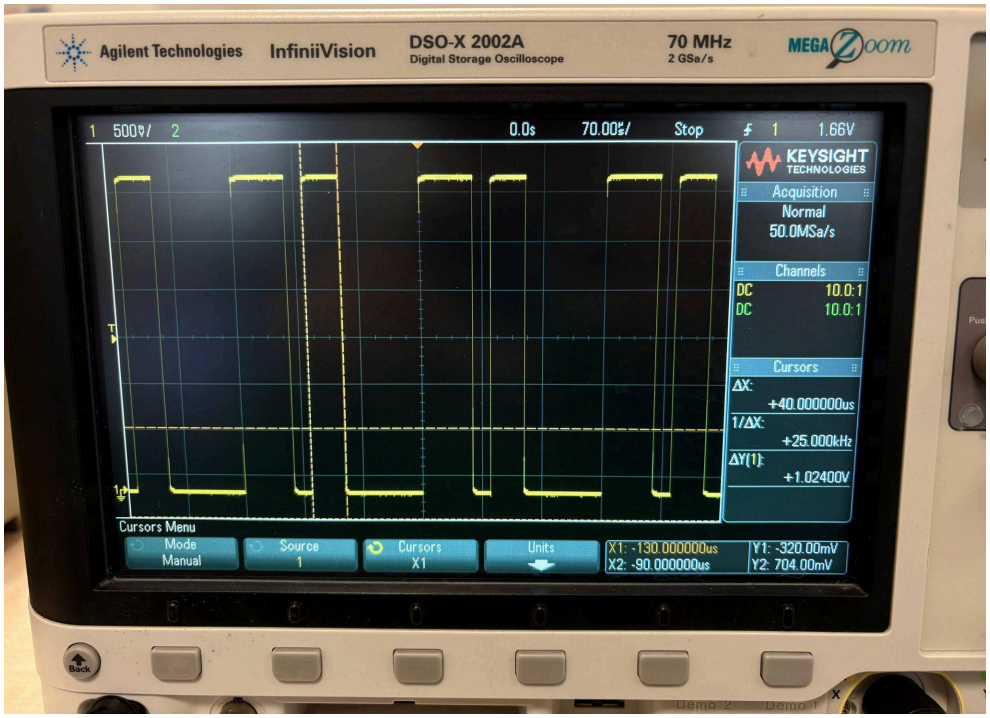


Figure 3: Oscilloscope capture of the waveform for sw="00" showing the full 210 µs period with cursor measurements showing $\Delta time$.

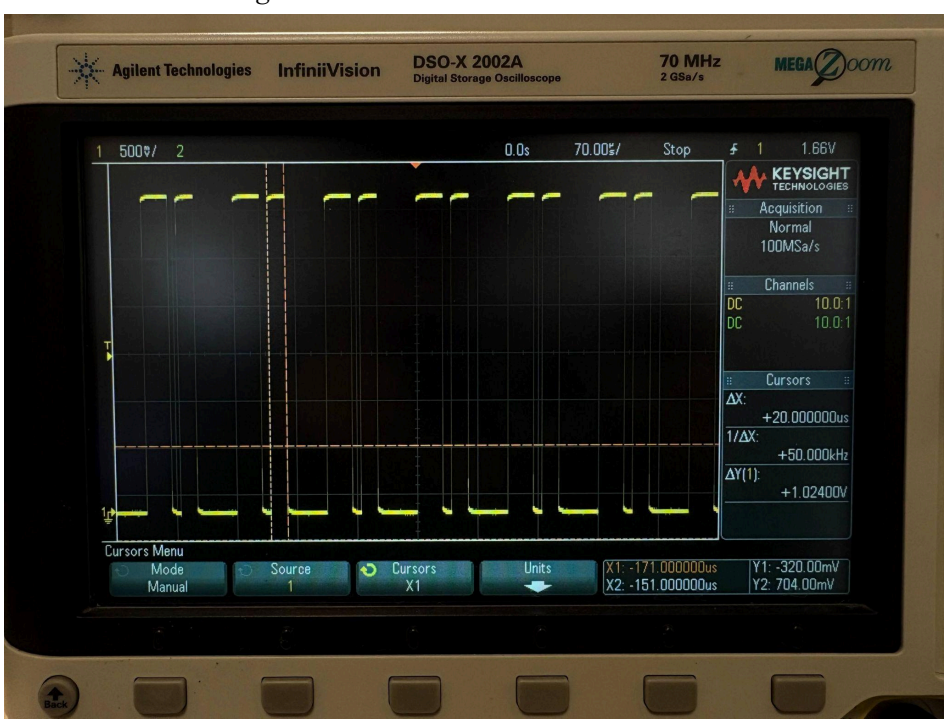


Figure 4: Oscilloscope capture of the waveform for sw="10" showing the compressed 105 µs period.

5. Design Considerations

During the development and testing process, several design decisions and observations deserve brief discussion as below:

Asynchronous Reset Warning: During block design validation, Vivado issued warning BD 41-1348, indicating that the reset pin of the waveform generator is connected to an asynchronous source while being associated with the Clocking Wizard output clock. In high-speed pipelined designs, an asynchronous reset can cause metastability if released near a clock edge. However, for this application, the reset originates from a manual push button whose assertion duration spans millions of clock cycles, making metastability practically impossible. A synchronous reset using a Processor System Reset IP would eliminate the warning but was deemed unnecessary for this design's operating conditions.

Clocking Wizard PLL Reset Isolation: The Clocking Wizard's reset input was deliberately tied to constant 0 rather than connected to the user push button. If the PLL were reset, its output clock would stop, and the waveform generator , which depends on that clock, would freeze entirely and could not recover until the PLL re-locks. By isolating the PLL reset, the push button only resets the waveform state machine while the clock infrastructure remains continuously operational.

Single-Process Architecture: The lab specification mandates that all logic reside within a single process with only the clock signal (and optional asynchronous reset) in the sensitivity list. This constraint ensures fully synchronous operation, eliminating the possibility of glitches on the output signal that could arise from combinational processes updating at different propagation delays. The output register s_wave is assigned inside the clocked process, guaranteeing that it transitions only on clock edges.

Port Name Verification: After generating the HDL wrapper from the block design, the auto-generated port names were cross-checked against the XDC constraints file. Vivado occasionally appends suffixes such as "_0" to external port names, which causes silent pin-mapping failures if the constraints file is not updated accordingly.

6. Conclusion

This laboratory session successfully demonstrated the design and implementation of a configurable arbitrary waveform generator on the Basys 3 FPGA. A four-phase finite state machine driven by a Clocking Wizard-generated 100 MHz clock produced precisely timed asymmetric pulse trains with zero quantization error. Real-time waveform modification was achieved through two physical switches controlling pattern selection and speed doubling, yielding four distinct output configurations. Both behavioral simulation and oscilloscope measurements confirmed exact agreement with theoretical timing calculations, validating the correctness of the counter arithmetic and state machine design.

Appendix A: VHDL Source Code

A.1. Main Module (waveform_gen.vhd)

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

entity waveform_gen is
   Port (
      s_clk    : in  STD_LOGIC;
      reset    : in  STD_LOGIC;
      sw       : in  STD_LOGIC_VECTOR(1 downto 0);
      wave_out : out STD_LOGIC;
      led      : out STD_LOGIC_VECTOR(1 downto 0)
   );
end waveform_gen;

architecture Behavioral of waveform_gen is

   constant PA_T1_1X : integer := 4000;
   constant PA_T2_1X : integer := 9000;
   constant PA_T3_1X : integer := 6000;
   constant PA_T4_1X : integer := 2000;

   constant PA_T1_2X : integer := 2000;
   constant PA_T2_2X : integer := 4500;
   constant PA_T3_2X : integer := 3000;
   constant PA_T4_2X : integer := 1000;

   constant PB_T1_1X : integer := 5500;
   constant PB_T2_1X : integer := 3500;
   constant PB_T3_1X : integer := 4500;
   constant PB_T4_1X : integer := 7500;

   constant PB_T1_2X : integer := 2750;
   constant PB_T2_2X : integer := 1750;
   constant PB_T3_2X : integer := 2250;
   constant PB_T4_2X : integer := 3750;

   signal s_phase_counter : integer range 0 to 9000 := 0;
   signal s_phase         : integer range 0 to 3 := 0;
   signal s_t1 : integer range 0 to 9000 := PA_T1_1X;
   signal s_t2 : integer range 0 to 9000 := PA_T2_1X;
   signal s_t3 : integer range 0 to 9000 := PA_T3_1X;
```

```
    signal s_t4 : integer range 0 to 9000 := PA_T4_1X;
    signal s_wave : std_logic := '0';

begin

    led <= sw;

    process(s_clk, reset)
    begin
        if reset = '1' then
            s_phase_counter <= 0;
            s_phase         <= 0;
            s_wave          <= '0';
            s_t1            <= PA_T1_1X;
            s_t2            <= PA_T2_1X;
            s_t3            <= PA_T3_1X;
            s_t4            <= PA_T4_1X;

        elsif rising_edge(s_clk) then

            case sw is
                when "00" =>
                    s_t1 <= PA_T1_1X;  s_t2 <= PA_T2_1X;
                    s_t3 <= PA_T3_1X;  s_t4 <= PA_T4_1X;
                when "01" =>
                    s_t1 <= PB_T1_1X;  s_t2 <= PB_T2_1X;
                    s_t3 <= PB_T3_1X;  s_t4 <= PB_T4_1X;
                when "10" =>
                    s_t1 <= PA_T1_2X;  s_t2 <= PA_T2_2X;
                    s_t3 <= PA_T3_2X;  s_t4 <= PA_T4_2X;
                when "11" =>
                    s_t1 <= PB_T1_2X;  s_t2 <= PB_T2_2X;
                    s_t3 <= PB_T3_2X;  s_t4 <= PB_T4_2X;
                when others =>
                    s_t1 <= PA_T1_1X;  s_t2 <= PA_T2_1X;
                    s_t3 <= PA_T3_1X;  s_t4 <= PA_T4_1X;
            end case;

            case s_phase is
                when 0 =>
                    s_wave <= '1';
                    if s_phase_counter = s_t1 - 1 then
                        s_phase_counter <= 0;
                        s_phase <= 1;
```

```
            else
                s_phase_counter <= s_phase_counter + 1;
            end if;
        when 1 =>
            s_wave <= '0';
            if s_phase_counter = s_t2 - 1 then
                s_phase_counter <= 0;
                s_phase <= 2;
            else
                s_phase_counter <= s_phase_counter + 1;
            end if;
        when 2 =>
            s_wave <= '1';
            if s_phase_counter = s_t3 - 1 then
                s_phase_counter <= 0;
                s_phase <= 3;
            else
                s_phase_counter <= s_phase_counter + 1;
            end if;
        when 3 =>
            s_wave <= '0';
            if s_phase_counter = s_t4 - 1 then
                s_phase_counter <= 0;
                s_phase <= 0;
            else
                s_phase_counter <= s_phase_counter + 1;
            end if;
        when others =>
            s_phase <= 0;
            s_phase_counter <= 0;
    end case;

    end if;
end process;

wave_out <= s_wave;

end Behavioral;
```

A.2. Behavioral Simulation Testbench (waveform_gen_tb.vhd)

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;
```

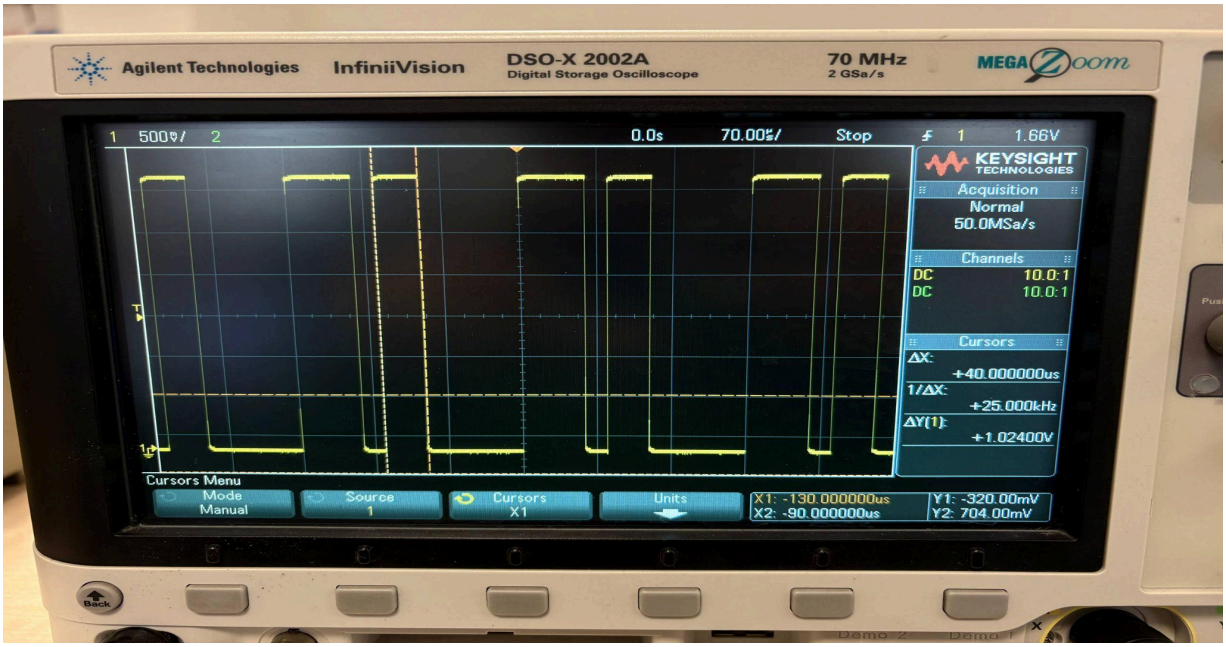
```
entity waveform_gen_tb is
end waveform_gen_tb;

architecture Behavioral of waveform_gen_tb is

    component waveform_gen
        Port (
            s_clk    : in  STD_LOGIC;
            reset    : in  STD_LOGIC;
            sw       : in  STD_LOGIC_VECTOR(1 downto 0);
            wave_out : out STD_LOGIC;
            led      : out STD_LOGIC_VECTOR(1 downto 0)
        );
    end component;

    signal tb_clk      : STD_LOGIC := '0';
    signal tb_reset    : STD_LOGIC := '0';
    signal tb_sw       : STD_LOGIC_VECTOR(1 downto 0) := "00";
    signal tb_wave_out : STD_LOGIC;
    signal tb_led      : STD_LOGIC_VECTOR(1 downto 0);

    constant CLK_PERIOD : time := 10 ns;
    signal sim_done     : boolean := false;

begin

    UUT: waveform_gen
        port map (
            s_clk    => tb_clk,
            reset    => tb_reset,
            sw       => tb_sw,
            wave_out => tb_wave_out,
            led      => tb_led
        );

    clk_proc: process
    begin
        while not sim_done loop
            tb_clk <= '0'; wait for CLK_PERIOD / 2;
            tb_clk <= '1'; wait for CLK_PERIOD / 2;
        end loop;
        wait;
    end process;
```

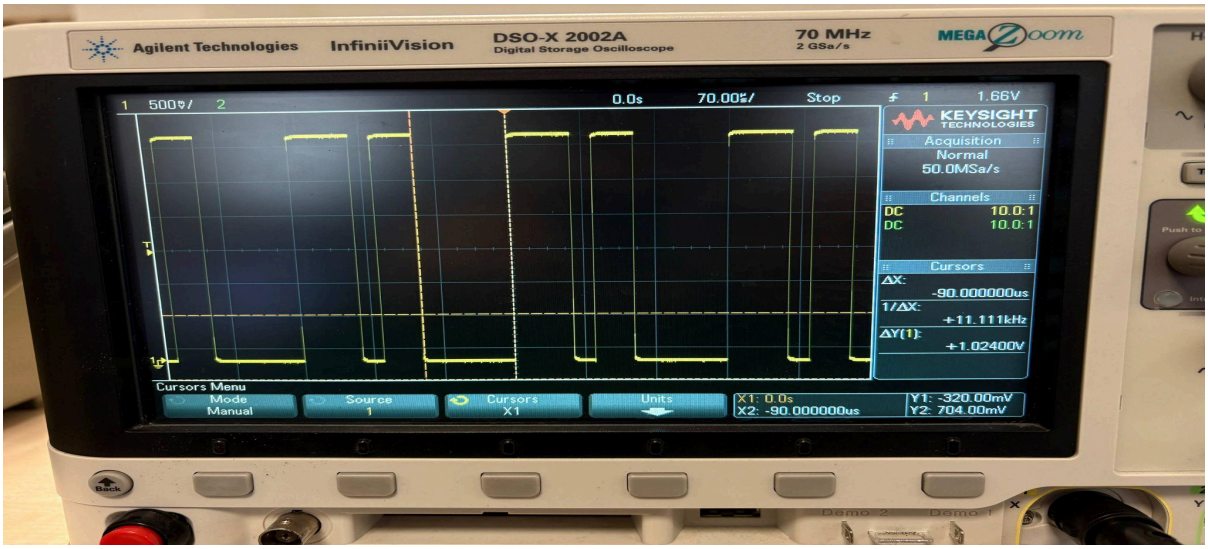
```
    stim_proc: process
    begin
        tb_reset <= '1';
        wait for 100 ns;
        tb_reset <= '0';
        wait for 50 ns;

        tb_sw <= "00";
        wait for 500 us;

        tb_sw <= "01";
        wait for 500 us;

        tb_sw <= "10";
        wait for 300 us;

        tb_sw <= "11";
        wait for 300 us;

        sim_done <= true;
        wait;
    end process;

end Behavioral;
```

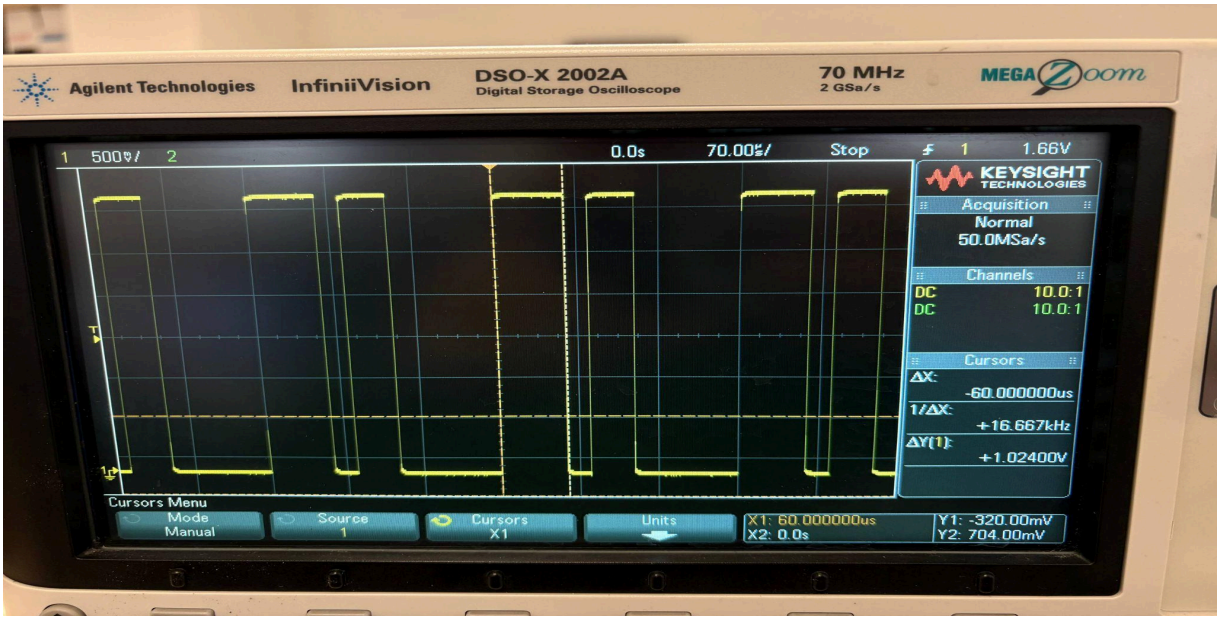
A.3. Constraints File (basys3_constraints.xdc)

```
set_property -dict { PACKAGE_PIN W5   IOSTANDARD LVCMOS33 } [get_ports sys_clk]
create_clock -add -name sys_clk -period 10.00 [get_ports sys_clk]

set_property -dict { PACKAGE_PIN U18  IOSTANDARD LVCMOS33 } [get_ports reset]

set_property -dict { PACKAGE_PIN V17  IOSTANDARD LVCMOS33 } [get_ports {sw[0]}]
set_property -dict { PACKAGE_PIN V16  IOSTANDARD LVCMOS33 } [get_ports {sw[1]}]

set_property -dict { PACKAGE_PIN J1   IOSTANDARD LVCMOS33 } [get_ports wave_out]

set_property -dict { PACKAGE_PIN U16  IOSTANDARD LVCMOS33 } [get_ports {led[0]}]
set_property -dict { PACKAGE_PIN E19  IOSTANDARD LVCMOS33 } [get_ports {led[1]}]

set_property CONFIG_VOLTAGE 3.3 [current_design]
set_property CFGBVS VCCO [current_design]
```

Appendix B: Additional Oscilloscope Verification Photographs







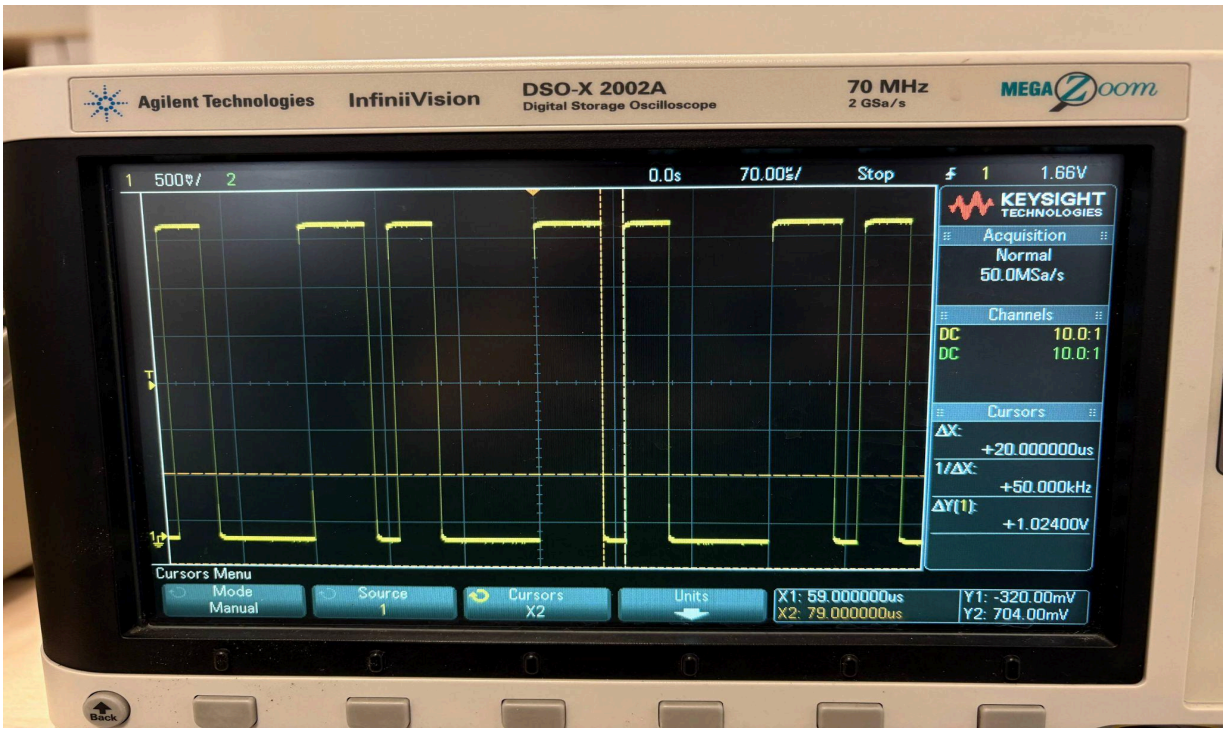
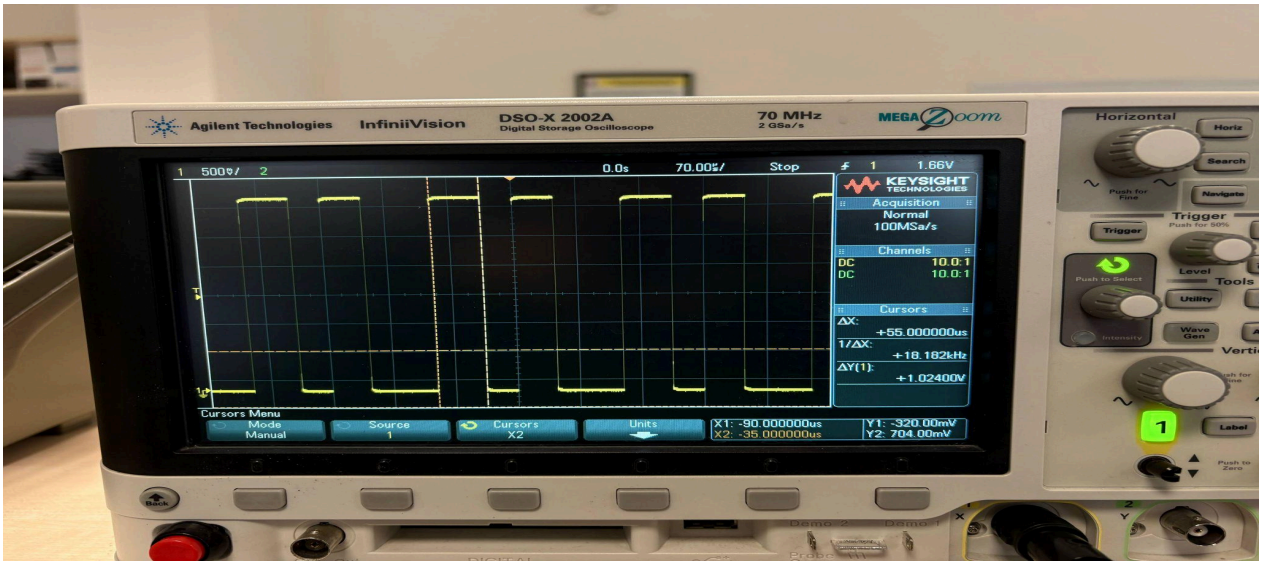


Figure-Table 1: Oscilloscope capture of Pattern A (sw="00") with cursor measurement showing time differences between phases.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InfiniiVision
DSO-X 2002A
Digital Storage Oscilloscope
70 MHz
2 GSa/s
MEGA Zoom
1 500mV/ 2
0.0s
70.00%/
Stop
1 1.66V
KEYSIGHT TECHNOLOGIES
Acquisition
Normal
100MSa/s
Channels
DC 10.0:1
DC 10.0:1
Cursors
ΔX: +55.000000us
1/ΔX: +18.182kHz
ΔY(1): +1.02400V
Cursors Menu
Mode Manual
Source 1
Cursors X2
Units
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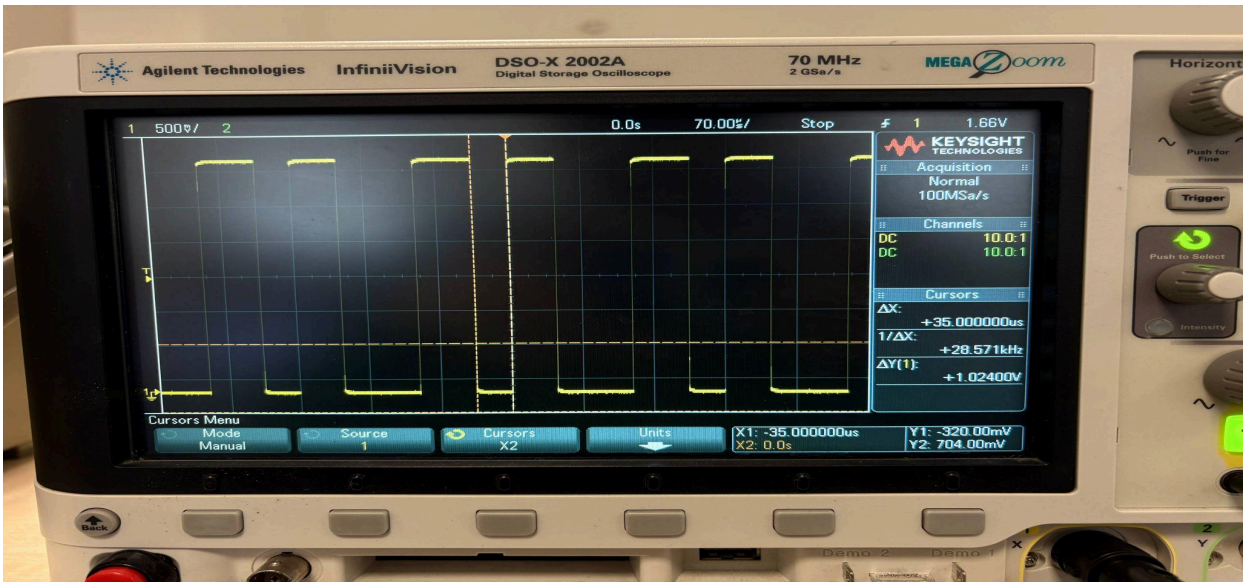
X2: -35.000000us
Y1: -320.00mV
Y2: 704.00mV
Horizontal
Trigger
Tools
Vertical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70.00%/
Stop
1 1.66V
KEYSIGHT TECHNOLOGIES
Acquisition
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Channels
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DC 10.0:1
Cursors
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1/ΔX: +28.571kHz
ΔY(1): +1.02400V
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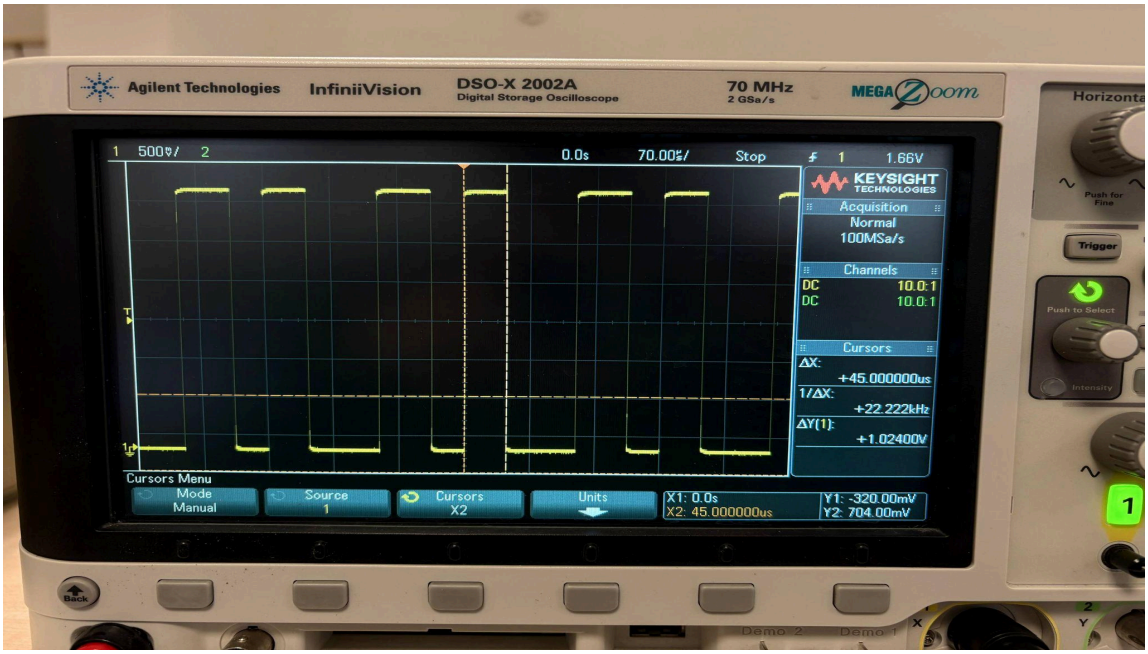
Y2: 704.00mV



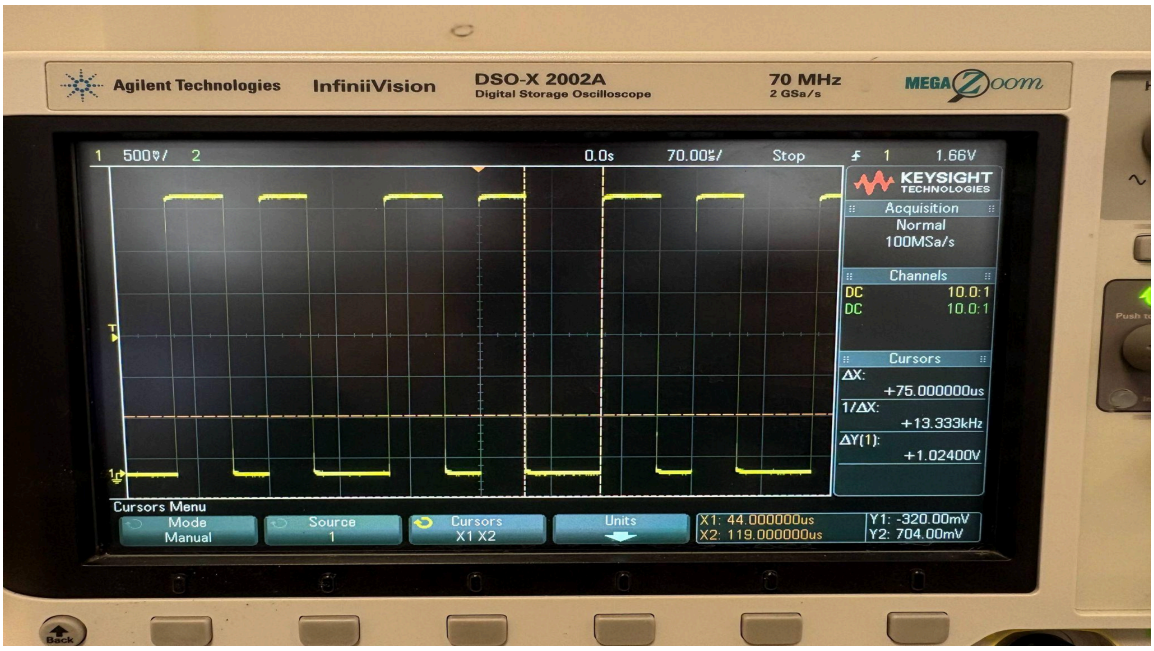


Figure-Table 2: Oscilloscope capture of Pattern A (sw="01") with cursor measurement showing time differences between phases.